Day 1: SAT/TSI Spiral Review – Exponents & Radicals

1. **SAT:** What is the value of $8^{2/3}$?

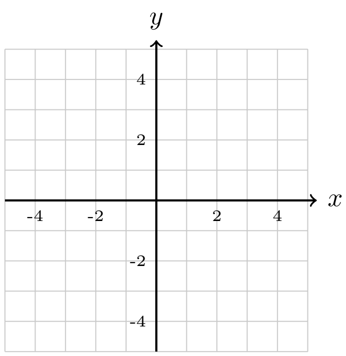
   A. 2 B. 4 C. $\dfrac{16}{3}$ D. 12

2. **TSI:** Simplify: $\dfrac{x^5 \cdot x^{-3}}{x^{-1}} =$ ____________

3. **SAT:** Which expression is equivalent to $\sqrt{x} \cdot x^2$?

   A. $x^{5/2}$ B. $x^{3/2}$ C. $x^{4/2}$ D. x^1

4. **Graphing Review:** Graph the line $y = 2x - 3$.



Slope (m): ________
y-intercept (b): ________
Plot at least 3 points!

Day 2: SAT/TSI Spiral Review – Polynomials & Factoring

1. **SAT:** Which is a factor of $x^3 + 27$?

   A. $(x - 3)$ B. $(x + 3)$ C. $(x + 9)$ D. $(x - 27)$

2. **TSI:** Factor completely: $x^3 - 4x^2 - 9x + 36 =$ ___________________________

   Hint: Group, then check for difference of squares!

3. **SAT:** If $f(x) = (x - 2)(x + 3)(x - 1)$, what are the zeros of f?

   A. $-2, 3, -1$ B. $2, -3, 1$ C. $2, 3, 1$ D. $-2, -3, -1$

4. **Graphing Review:** Find the slope of the line through $(1, 4)$ and $(3, 10)$.

   $m = \dfrac{y_2 - y_1}{x_2 - x_1} =$ ____________________

Day 3: Final Review – Mixed SAT/TSI

1. **SAT:** If $a^2 = 7$, what is a^4?

   A. 14 B. 28 C. 49 D. 2401

2. **TSI:** Simplify: $(3x - 2)(x^2 + 4x - 1) =$ _________________________________

3. **SAT:** Use synthetic division to find the remainder when $x^3 + 2x^2 - 5x + 1$ is divided by $(x - 2)$.

$$\begin{array}{c|cccc} 2 & 1 & 2 & -5 & 1 \\ \hline & & & & \end{array}$$

Remainder: ___________

4. **Graphing Review:** Write the equation of a line with slope $-\frac{1}{2}$ passing through $(4, 1)$.

   Using point-slope: $y - y_1 = m(x - x_1)$ Equation: ____________________

Key Formulas to Remember: $a^m \cdot a^n = a^{m+n}$ $\frac{a^m}{a^n} = a^{m-n}$ $(a^m)^n = a^{mn}$ $a^{m/n} = \sqrt[n]{a^m}$ $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$

Answer Key: D1: 1. B 2. x^3 3. A 4. $m = 2, b = -3$ **D2:** 1. B 2. $(x-4)(x-3)(x+3)$ 3. B 4. $m = 3$
D3: 1. C 2. $3x^3 + 10x^2 - 11x + 2$ 3. R=7 4. $y = -\frac{1}{2}x + 3$ or $y - 1 = -\frac{1}{2}(x - 4)$